

# Implementation of chaotic frequency modulation based spread spectrum communication system in software-defined radio

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**Abstract**—This paper is devoted to the software-defined radio (SDR) implementation of frequency modulated antipodal chaos shift keying (FM-ACSK) communication system. This novel and perspective class of spread-spectrum systems employs chaotic synchronization for the acquisition and tracking of the spreading signal and does not need resource-demanding cross-correlation usually employed for this purpose. This paper aims to present how to implement a full FM-ACSK communication system, including additional synchronization circuitry, using SDR and MATLAB Simulink. Moreover, for the first time, the performance of FM-ACSK in real conditions using over-the-air transmission between two SDRs is assessed. In the paper, it is shown that the presented class of the spread spectrum systems demonstrates good performance in low signal-to-noise ratio (SNR) conditions, and it is immune to large carrier frequency offset (CFO). The proposed solution fits well to low-speed long-range wireless sensor network (WSN) applications.

**Index Terms**—Software-defined radio, chaotic communication, spread spectrum communication, frequency modulation, communication systems, Matlab, physical layer

## I. INTRODUCTION

Nowadays, transmission security is an essential topic for the increasing use of data communications. A common way to ensure data security nowadays is data encryption, ensuring that data is invalid in the hands of malicious users, as it cannot be decrypted without the necessary cipher key. Although much of the data in the modern technological era is protected this way, this encryption implementation is not always perfect, as it inadvertently leaves vulnerabilities. These vulnerabilities can be reduced by the introduction of security at the lowest layers of the communication system.

Employment of the chaotic behavior of the dynamical systems paves the way for creating secure and hidden communication systems. Such solutions, in general, can be characterized by a wide frequency spectrum, and their waveform can look like noise to the external observer. Given characteristics of these analog signals can also be employed in digital communications. Comprehensive reviews of these methods are available in textbook [1] and review paper [2]. New single-carrier [3] and multi-carrier chaotic communication techniques continue to emerge [4].

This paper is devoted to software-defined radio (SDR) implementation of a spread-spectrum system where the chaos generator generates the continuous spreading waveform, and spreading is performed in the frequency domain using frequency modulation (FM). Such systems have been known for a long time, and variations of this approach are reported in publications. In one of the first publications [5], authors propose a Bernoulli map for the generation of the chaotic frequency hopping (FH) sequence and present system implementation in fixed digital logic. In another publication, [6] entirely analog implementation of discrete-time chaotic generator with FM is reported. In publication [7], authors propose to use phase locked loop (PLL) for the FM demodulation in binary frequency shift keying (BFSK) system with chaotic spreading. In publication [8], authors provide a detailed analysis of an improved high-speed frequency modulated differential chaos shift keying (FM-DCSK) system, which uses discrete-time chaotic sequences and selective orthogonalization. Unlike previous schemes, the despreading in this modulation scheme is performed using a classical cross-correlation approach. In a recent publication, [9] authors propose to use chaos-based frequency modulation for simultaneous sensing and data communication.

The analog nature of the chaotic dynamics requires a careful approach to implementing such systems because they are susceptible to the parameters. The digital communication system's physical layer and chaotic generators can be implemented using analog circuitry [6], [10]. Alternatively, the chaotic waveforms and sequences can be generated by digital schemes. This approach is especially convenient for discrete-time chaotic systems generating chaotic sequences [5], [11]. However, the quantization of the values by digital circuits will inevitably lead to the system's periodic behavior, and the number of bits employed to represent the values must be as large as possible. Early digital chaotic generators were implemented using fixed logic [5], however, since modern system-on-chips (SoCs) and field-programmable gate arrays (FPGAs) became available, the chaotic generators can also be programmed and run on an FPGA [12] or general-purpose

computer connected to SDR [13], [14], [11].

This paper, for the first time, presents the implementation and assesses the performance of a full of frequency modulated antipodal chaos shift keying (FM-ACSK) transceiver, which includes up/down-conversion, pulse shaping, and symbol synchronization. Moreover, the performance of the given prototype is assessed in an industrially relevant environment to determine its suitability to wireless sensor network (WSN) applications.

This paper is organized as follows: Section II is devoted to the description, and related theoretical background of the chaotic subsystem of the communication system, Section III is committed to the SDR implementation aspects, Section IV pays attention to the experimental verification of the communication system prototype. Finally, Section V explains the obtained results and concludes the paper.

## II. DESCRIPTION OF FM-ACSK SYSTEM

FM-ACSK communication system is based on chaotic synchronization between master and slave chaos generators. This section describes the mathematical principles of chaos generation and synchronization and the two-step modulation used in the communication system under study, which was first presented in [15].

### A. Chaos generator and synchronization

Fourth-order modified Chua's circuit chaos generator was used as a base of the communication system. It can be described by a system of four ordinary differential equations:

$$\begin{cases} \frac{dp_1}{dt} = -g(p_1, p_3)(p_1 - p_3) - p_2 \\ \frac{dp_2}{dt} = p_1 + \gamma p_2 \\ \frac{dp_3}{dt} = \theta(g(p_1, p_3)(p_1 - p_3) - p_4) \\ \frac{dp_4}{dt} = \sigma p_3 \end{cases} \quad (1)$$

where  $p_1, p_2, p_3, p_4$  are system's state variables,  $\gamma, \theta, \sigma$  are system coefficients, but  $g(p_1, p_3)$  is piecewise linear function, described as:

$$g(p_1, p_3) = \begin{cases} c(p_1 - p_3 - d) & (p_1 - p_3) > d \\ 0 & (p_1 - p_3) \leq d \end{cases} \quad (2)$$

where  $c = 3$  and  $d = 1$  are system coefficients that define the chaotic behavior of the dynamical system.

The output signal of the chaos generator ( $R_{out}$ ) consists of the weighted sum of the system's state variables and the piecewise linear function:

$$R_{out} = p_1 k_1 + p_2 k_2 + p_3 k_3 + p_4 k_4 + g(p_1, p_3) \quad (3)$$

where weight coefficients  $k_1, k_2, k_3$ , and  $k_4$  define the output signal form variation, which can be used for multiple access implementation purposes. System and weight coefficient values are shown in Table I.

TABLE I  
COEFFICIENTS OF DIFFERENTIAL EQUATIONS.

| $\gamma$ | $\theta$ | $\sigma$ | $c$ | $d$ | $k_1$   | $k_2$   | $k_3$ | $k_4$  |
|----------|----------|----------|-----|-----|---------|---------|-------|--------|
| 0.5      | 10       | 1.5      | 3   | 1   | -2.6302 | -0.6054 | 0.587 | 0.7763 |

Chaotic synchronization is employed to estimate the received binary symbol value. Chaotic synchronization recreates the master chaos generator's state variables in the slave chaos generator, which is done by the linear error feedback method. The main distinctive feature of the slave chaos generator compared to the master is that it doesn't have its own piecewise linear function. This function gets restored by subtracting the weighted sum of the state variables from the input signal  $R_{in}$ :

$$g'(p_1, p_3) = R_{in} - p_1' k_1 + p_2' k_2 + p_3' k_3 + p_4' k_4 \quad (4)$$

where  $p_1', p_2', p_3', p_4'$  are state variables of the slave chaos generator,  $g'(p_1, p_3)$  is the restored piecewise linear function, and  $R_{in}$  is the chaotic input signal, which in the simplified case is equal to  $R_{out}$ , but in the practical system, it is received from the FM demodulator with some additional distortions caused by the transmission channel noises and FM.

Chaotic synchronization is established when the difference between master and slave state variables (considering the transmission delay) reaches zero. To evaluate the synchronization error, the slave chaos generator's output signal  $R'$  is subtracted from its input  $R_{in}$ :

$$e = R' - R_{in} \quad (5)$$

The output signal  $R'$  is formed in a similar manner as in the master chaos generator, but using an absolute value of piecewise linear function  $g'$ :

$$R' = p_1' k_1 + p_2' k_2 + p_3' k_3 + p_4' k_4 + |g'(p_1, p_3)| \quad (6)$$

According to (2), piecewise linear function  $g(p_1, p_3)$  values are always positive. Therefore, if recovered function  $g'$  values are in the negative range, they are recovered incorrectly due to the lack of chaotic synchronization. Thus,  $|g'(p_1, p_3)|$  is employed to estimate synchronization error. However, the obtained synchronization error is inconsistent since the piecewise linear function at some moments is equal to zero. To ensure consistent and gradual synchronization error estimation, the root mean square (RMS) of it is calculated.

### B. FM-ACSK modulation

The modulation in the present communication system consists of two layers – antipodal chaos shift keying (ACSK) modulation and the FM (see Fig.1) on top of it. ACSK layer uses three chaos generators – one at the transmitter and two at the receiver. At the transmitter, the master chaos generator's output chaotic signal gets passed forward without changes if data bit "0" is transmitted, or it gets inverted when data bit "1" is transmitted. At the receiver, there are two slave chaos generators. The received chaotic signal gets applied without changes to one of the generators, whereas to the other

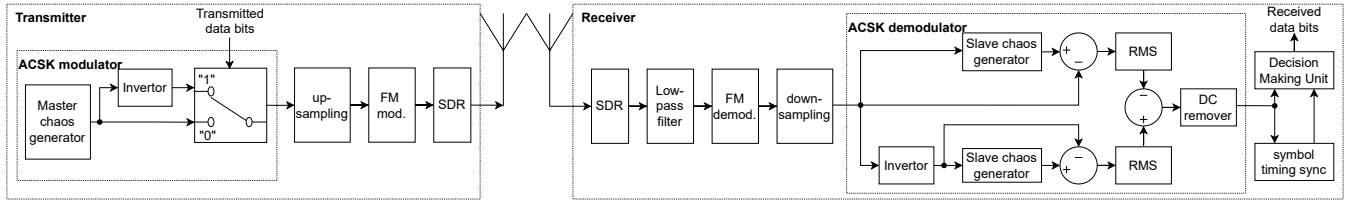


Fig. 1. Block diagram of FM-ACSK communication system SDR model.

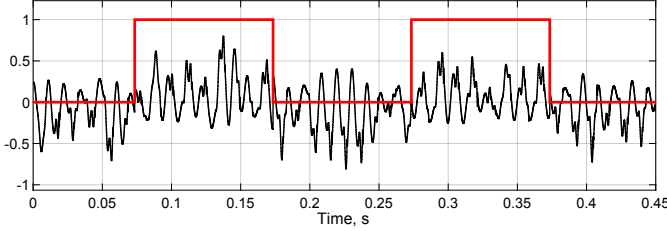


Fig. 2. Spreading of the bits using continuous chaotic waveform. Data bits are drawn with the red line, switched chaotic waveform – black line. There are approximately 9 oscillations per bit.

generator inverted version is passed. This way, depending on the use of the signal inversion at the transmitter, the received chaotic signal will not be inverted only at one of the two slave chaos generators, which will provide a much smaller estimated synchronization error compared to another generator. RMS values of both receiver generators' synchronization errors are compared with each other to decide what data bit value was transmitted.

It is essential to understand that ACSK demodulator performs estimation of the transmitted bit by analyzing a relatively long piece of the chaotic signal (see Fig. 2). Therefore, it can be said that ACSK demodulator performs despreading of the received signal by means of chaotic synchronization. In classical spread spectrum (SS) systems, despreading is performed by means of a computationally demanding correlator. Employing the chaotic synchronization instead of correlation saves computation resources, and ACSK can be seen as a novel type of SS modulation. More details about ACSK can be found in [16].

On top of ACSK modulation, FM is performed for several purposes. With FM modulation, the transmitted chaotic signal can be spread over the broader frequency range, providing extra diversity and increasing immunity to the interference. Moreover, the transmitted FM modulated signal has a constant envelope, which allows using energy-efficient power amplifier (PA) with small backoff. Finally, employment of non-coherent FM significantly lowers requirements for the stability of the carrier frequency and increases immunity to carrier frequency offset (CFO), which is the main hurdle in coherent modulations.

### III. SDR IMPLEMENTATION

For the testing of the proposed modulation technique in real conditions, it was decided to create SDR-based prototype. The

hardware part of the prototype consists of two Adalm Pluto<sup>TM</sup> SDR connected to one or two computers. The software part of the prototype consists of transmitter and receiver models implemented in MATLAB Simulink. SDR software is developed from the Simulink models employed in research [15] by adding timing synchronization and SDR-related processing. The parameters of the prototype are provided in Table II.

#### A. FM-ACSK transmitter

In accordance with Fig. 1 the transmitter consists of ACSK modulator, which is connected to the baseband FM modulator unit via up-sampling unit, for conversion of ACSK modulator sample rate into SDR baseband sample rate. FM modulator works at SDR baseband sample rate because such setup allows using large FM deviations. The antipodal shift keying is performed by switching between the direct or inverted signals from the chaotic generator. The output of FM modulator is connected to the Adalm Pluto SDR driver.

#### B. Master chaos generator in the transmitter

The chaotic generator in the ACSK modulator in the transmitter is built in accordance with principles described previously in Section II-A. To use SDR the model must run in discrete time. Therefore, it was necessary to discretize the continuous-time model and replace differential equations with difference equations. In terms of the Simulink implementation, it means the employment of "Discrete State-Space" systems described by the equation:

$$\begin{aligned} x(n+1) &= Ax(n) + Bu(n) \\ y(n) &= Cx(n) + Du(n) \end{aligned} \quad (7)$$

where in  $u$  is the input,  $x$  is the state, and  $y$  is the output. In our case, all "Discrete State-Space" systems are integrators with the same parameters  $A = 1$ ,  $B = 0.05$ ,  $C = 1$ ,  $D = 0$ . Therefore, the equation (7) can be rewritten as:

$$\begin{aligned} x(n+1) &= x(n) + 0.05u(n) \\ y(n) &= x(n) \end{aligned} \quad (8)$$

The Simulink design of the chaotic generator in the transmitter is depicted in Fig. 3.

TABLE II  
PARAMETERS OF A COMMUNICATION SYSTEM PROTOTYPE.

| Carrier frequency | SDR baseband sample rate | ACSK sample rate | Samples per bit | Data rate |
|-------------------|--------------------------|------------------|-----------------|-----------|
| 867.4 MHz         | 240 kHz                  | 10 kHz           | 1000            | 10 bit/s  |

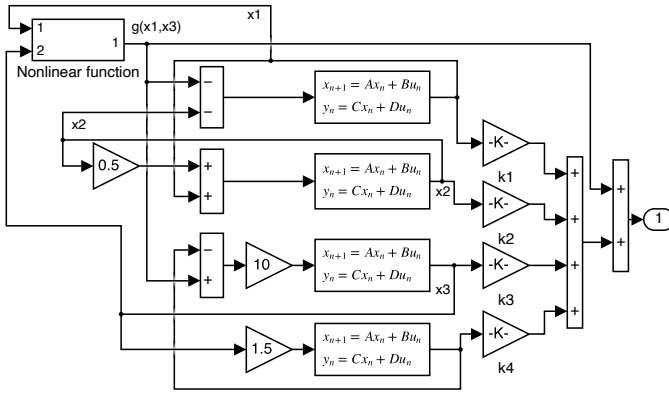


Fig. 3. Implementation of chaotic generator in the ACSK modulator.

### C. FM-ACSK receiver

In the receiver part, see Fig. 1, the key blocks are FM demodulator, ACSK demodulator, and symbol timing synchronization unit. Additionally, there is low-pass filter (LPF) at the input of the receiver for the limiting noise in cases when useful bandwidth is much smaller than the baseband rate of SDR. Downsampler is necessary for the aligning sample rates of SDR and ACSK demodulator, which works at a much lower sample rate. FM-ACSK demodulator performs despreading of the chaotic waveforms through chaotic synchronization as described in detail in Section II-B.

For decision-making, two different approaches are tested. The decisions can be made by the classic strobe-based detector (SD) by sampling waveform at the output of ACSK demodulator at the optimal time instant. However, an energy-based detector (ED), which calculates the mean value of the whole symbol interval, provides better results (see Fig. 7). Moreover, the system is less sensitive to symbol timing offset (STO) in the latter case. Decision-making is governed by a symbol timing synchronization block, which ensures decision-making at optimal time instants.

#### D. Slave chaos generators in the receiver

FM-ACSK demodulator in the receiver contains two identical chaotic generators which can synchronize either to direct or inverted chaotic waveform. Those generators are very similar to the generator employed in the transmitter (see Section III-B), except that the nonlinear function is reconstructed from the incoming signal. The mechanism of how it happens in detail was described in Section II-A. Simulink design of the receiver's chaotic generator is depicted in Fig. 4.

### E. Symbol timing synchronization

Symbol timing synchronization is a crucial task that has to be solved in any data transmission system. The block diagram of the timing synchronization unit employed in the proposed FM-ACSK system is shown in Fig. 5.

The RMS unit outputs a gradually increasing signal in case if respective slave chaos generator is not synchronized, whereas the RMS unit of the synchronized generator outputs

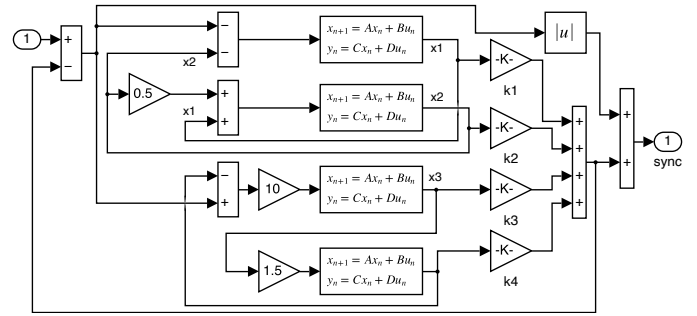


Fig. 4. Implementation of the chaotic generator and synchronization circuit in the ACSK demodulator.

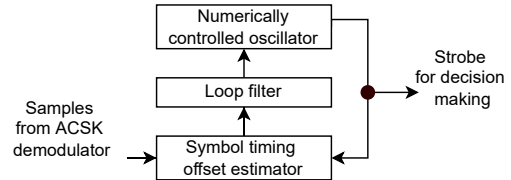


Fig. 5. Block diagram of symbol timing synchronization unit.

approximately zero. The bit decisions must be made from the summary waveform produced by subtracting signals from the RMS units. The optimal decision point is located at the time instant when the summary waveform has reached the positive or negative peak. The communication system is strongly susceptible to the STO as the value at the input of the decision device changes rapidly, i.e., the eye diagram is narrow.

The processing in the symbol timing synchronization unit resembles a classic loop scheme with a data-aided estimator and numerically controlled oscillator (NCO). The estimator looks for the peaks in the incoming signal and measures their offset from the midpoint of the time window. Via the loop filter, which averages the offsets, the position is sent to the NCO, which adjusts the timing window so that the offset gradually decreases.

## IV. EXPERIMENTAL RESULTS

A test and measurement campaign was carried out to verify the performance of the created SDR-based communication prototype. In the first test, two Adalm Pluto<sup>TM</sup> units were connected to the same computer, and a single Simulink model performed both reception and transmission. This setup aims to assess the prototype's performance in case of ideal timing synchronization. For the transmission and reception, identical 27 cm long coil whip 868 MHz antennas located 5 m apart were used.

#### A. BER versus transmit power

For the measurement of bit error ratio (BER), the gain of the receiver was kept at the levels 10 dB and 20 dB, whereas transmitter power was varied in range  $-42 - -22$  dBm corresponding to the Adalm Pluto output power  $0.006 - 6$   $\mu$ W. Results depicted in Fig. 7 show that in all cases, an observable

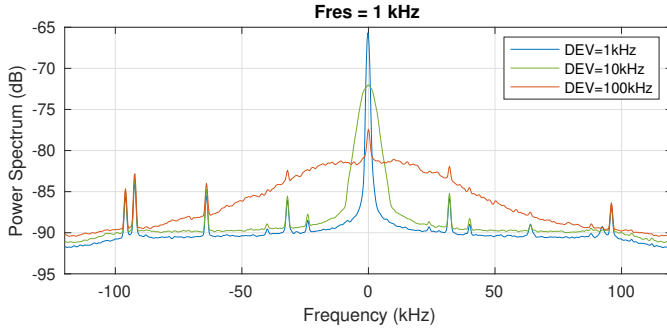


Fig. 6. Power spectra in the receiver model after SDR block. Transmit gain -50 dB (power -42 dBm, 63 nW) and receiver gain 20 dB. Frequency resolution is 1000 Hz. DEV stands for FM deviation.

threshold effect could be related to the employment of FM. The energy-based detector allows for achieving much better results because it averages ACSK demodulator output samples over the whole symbol interval.

It is also noticeable that smaller FM deviation allows achieving lower BER. This happens because a larger deviation needs wider frequency bandwidth of the input low-pass filter, which inevitably leads to lower signal-to-noise ratio (SNR) after the filter.

The power spectrum of the received signals is shown in Fig. 6. the SNR of the received signals can be estimated from the power spectrum using the approximate formula:

$$SNR = G_s - G_n + 10 \log_{10}(K_s) - 10 \log_{10}(K_n) \quad (9)$$

where  $G$  is power spectral density reading from the power spectrum and  $K$  is a count of frequency bins occupied by the signal (subscript s) or noise (subscript n). Using this approximate formula, we can estimate that SNR of signals whose spectra are shown in Fig. 6 is about 10 dB.

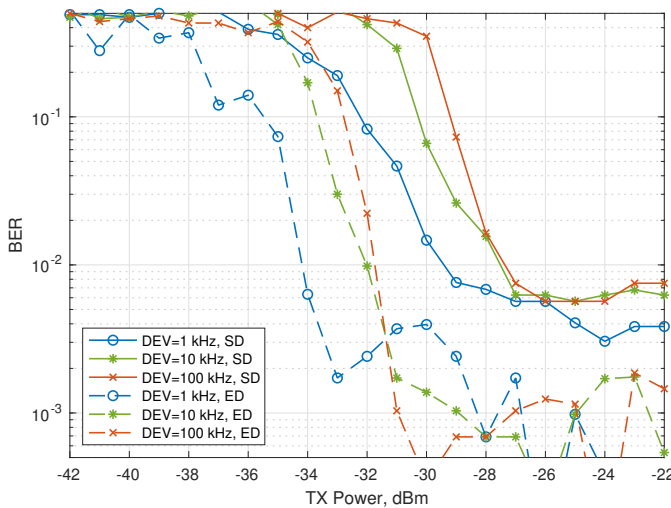


Fig. 7. BER versus transmit power in a single model scenario with different bit detectors. DEV stands for FM deviation, SD means strobe-based detector, and ED – energy-based detector. Receiver gain is 10 dB.

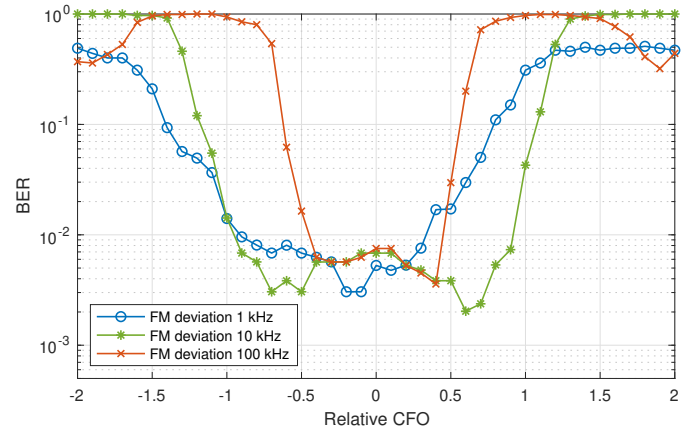


Fig. 8. BER versus relative CFO in the single model scenario and strobe-based bit detector. Absolute CFO can be calculated by multiplying relative CFO by the deviation. The receiver gain is 10 dB.

### B. BER versus CFO measurement

In the second test impact of the CFO on the quality of the communication was measured. For this purpose, the receiver has been detuned from the transmitter frequency, and BER has been counted. Results of this experiment depicted in Fig.8 show that system is tolerant to huge CFO (at least half of the deviation), and an increase of FM deviation also increases the communication system's tolerance to the CFO. Hypothetically, the shape of those curves is primarily determined by the low-pass filter at the input of the baseband receiver. A broader filter would allow tolerating even larger CFO at the cost of decreased SNR.

### C. Communication between two computers

In the ultimate test, the SDRs have been connected to two separate computers. The distance and other parameters have

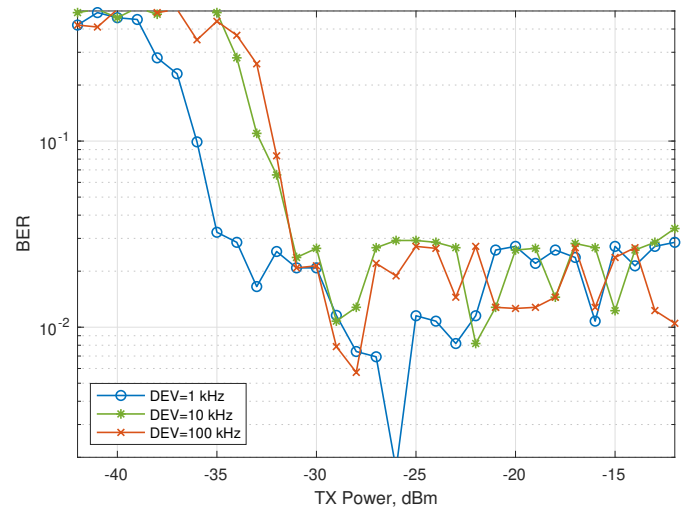


Fig. 9. BER versus transmit power using separate transmitter and receiver on different computers connected to SDR. DEV stands for FM deviation. Receiver gain is 10 dB. The strobe-based detector is employed.



been maintained similarly as in Section IV-A.

To synchronize Bernoulli generators in the receiver and the transmitter models, a network connection between the computers running SDR models is employed. The transmitter model has a "TCP/IP Client Send" Simulink block, whereas the receiver has a "TCP/IP Client Receive." Moreover, on the receiver computer, there is a separate TCP/IP server application, which forwards all requests from one client to another. When FM-ACSK transmitter model starts, it sends a request to the server, which forwards this request to the client on the receiver side. The receiver model does not start the Bernoulli generator unless it receives the request from the transmitter. Considering the low speed of the transmission, 2 – 5 ms latency of the TCP/IP network does not affect the synchronization accuracy.

The achieved BER in two computer scenario is depicted in Fig. 9. The calculations have been performed with 60 s delay to eliminate errors caused by the timing synchronization process. By comparing Fig. 7 and Fig. 9 we can conclude that the symbol timing synchronization process introduces bit errors which lead to error floor at approximately  $10^{-2}$ . However, the achieved BER is sufficient for the employment of commonly used error correction code (ECC) algorithms, such as low-density parity-check (LDPC) or Turbo codes.

## V. CONCLUSIONS

The purpose of the given research was to create a real-world SDR-based prototype for the testing, tuning, and assessing the performance of FM-ACSK communication system. The work included creating Simulink software for SDR, tuning the parameters, and testing.

The experiments have shown that the proposed communication system can provide acceptable BER at SNR 0 – 10 dB and is comparable with digital SS systems with a spreading factor around 10. The high sensitivity of the system is achieved through the processing gain caused by spreading the data bits over many cycles of the chaotic waveform. The proposed modulation does not need equalization and can work with multi-path communication channels. In conjunction with forward error correction (FEC), it is suitable for employment in WSNs, where many nodes communicate simultaneously.

Moreover, the system maintains acceptable BER at CFO, which is equal to or less than 50% of FM deviation. Such characteristics are achieved by employing non-coherent FM on top of coherent ACSK.

The proposed communication system needs to employ a high upsampling factor. The broad spectrum of chaotic oscillations requires approximately 110 samples per chaotic cycle, which leads to 1000 samples per one bit. This setting inevitably leads to low transfer speed and high spectrum occupancy even at low FM deviations and small FM modulation indexes. This can be considered one of the limitations of the given modulation scheme, and it can be solved using faster hardware. For example, using FPGA-based SDR implementations of ACSK similar to [12], it could be possible to achieve much higher transfer rates.

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